

Problem

How much inductance is needed to resonate at 5 kHz with a capacitance of 12 nF?

- (a) 2,652 H
- (b) 11.844 H
- (c) 3.333 H
- (d) 84.43 mH

Step-by-step solution

Step 1 of 1

We know the resonant frequency $\omega_o = \frac{1}{\sqrt{LC}}$ —(1)

Where $L \rightarrow$ Inductance of the circuit.

$C \rightarrow$ Capacitance of the circuit = 12 nF

We also know,

$$\omega_o = 2\pi v \quad \text{---(2)}$$

$V \rightarrow$ Frequency of the circuit = 5 kHz

From (2)

$$\omega_o = 2\pi \times 5 \times 10^3 = 31415.92 \text{ rad/s}$$

$$\text{From (1)} \quad \omega_o^2 = \frac{1}{LC}$$

$$\begin{aligned} \Rightarrow L &= \frac{1}{\omega_o^2 \times C} = \frac{1}{(31415.92)^2 \times 12 \times 10^{-9}} \\ &= 8.443 \times 10^{-11} \times 10^9 \\ &= 8.443 \times 10^{-2} \text{ H} \end{aligned}$$

$$\boxed{L = 84.43 \text{ mH}}$$